

Goodness-of-Fit Tests via Kernel Density Estimation: Univariate and Multivariate Approaches

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Abstract

This paper develops new goodness-of-fit tests based on kernel density estimation (KDE) methods for both univariate and multivariate settings. The proposed test statistics compare a parametric null model with a nonparametric KDE-based estimator under the alternative. For univariate densities with bounded support, we apply boundary correction using the reflection method to mitigate edge bias. In the multivariate setting, we account for bandwidth matrix selection and geometric invariance. Theoretical results demonstrate that the proposed tests are consistent and asymptotically converge to the Kullback–Leibler divergence under fixed alternatives. Extensive simulations and real data applications confirm the strong empirical performance of the tests, particularly in detecting smooth deviations from the null distribution.

Keywords: Bootstrap; Boundary correction; Goodness-of-fit test; Kernel density estimation; Multivariate density estimation.

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1 Introduction

Assessing the adequacy of a parametric statistical model is a fundamental task in statistical inference, with widespread applications in model selection, diagnostics, and validation. Classical approaches to goodness-of-fit testing, such as the Kolmogorov–Smirnov and Anderson–Darling tests, evaluate the plausibility of a hypothesized model by measuring the discrepancy between the empirical cumulative distribution function and the cumulative distribution function under the null hypothesis. These methods are well established and widely used when the null distribution is fully specified or involves only a small number of parameters (D’Agostino and Stephens, 1996).

Kernel density estimation (KDE) provides a rigorous foundation for goodness-of-fit tests in both univariate and multivariate settings. Bickel and Rosenblatt (1973) pioneered this approach by establishing global deviation measures through supremum-norm comparisons between kernel density estimates and true densities. Fan (1994) expanded the framework by developing L^2 -distance tests with optimal bandwidth selection to enhance power against local alternatives. Subsequent work by Fan (1998) introduced significance tests using smoothed parametric densities to reduce specification bias in null models. For practical implementation, Pavia (2015) operationalized Fan’s (1994) methodology in the GoFKernel R package. In multivariate contexts, Duong and Hazelton (2003) derived data-driven bandwidth matrix selection techniques that maintain asymptotic consistency of KDE-based tests in higher dimensions. Collectively, these contributions establish KDE as a versatile nonparametric framework for robust model validation beyond classical parametric assumptions.

Let $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n) \in \mathbb{R}^d$ be independent and identically distributed (i.i.d.) observations from a d -dimensional continuous distribution with unknown density f . Each $\mathbf{X}_i \in \mathbb{R}^d$ is typically treated as a row vector: $\mathbf{X}_i = (X_{i1}, X_{i2}, \dots, X_{id})$, where X_{ij} denotes the j -th coordinate of the i -th observation. Throughout this paper, when $d = 1$, i.e., in the univariate case, we drop the bold

notation and write $X_1, \dots, X_n \in \mathbb{R}$ for simplicity. We consider testing the null hypothesis

$$H_0 : f = f_0(\cdot; \theta), \quad \theta \in \Theta, \quad (1)$$

where f_0 is a known parametric family and θ is an unknown parameter vector. The alternative hypothesis is the general nonparametric alternative $H_1 : f \neq f_0(\cdot; \theta)$ for all $\theta \in \Theta$, where $\Theta \subset \mathbb{R}^p$. Our approach constructs test statistics by comparing the fitted null density $f_0(\mathbf{x}; \hat{\theta})$ to a kernel-based estimator $\hat{f}_n(\mathbf{x})$ of the true density. Specifically, we focus on statistics of the form

$$T_n = \prod_{i=1}^n \frac{\hat{f}_n(\mathbf{X}_i)}{f_0(\mathbf{X}_i; \hat{\theta})}, \quad (2)$$

where $\hat{\theta}$ denotes the maximum likelihood estimator of θ under the null model. The statistic T_n may be viewed as a nonparametric analogue of the likelihood ratio statistic. Large values of T_n indicate that the empirical distribution differs substantially from the model under H_0 . Thus, the proposed approach compares a nonparametric kernel estimate of the density with a parametric model specified under the null hypothesis. The resulting test statistic takes the form of a likelihood ratio-type criterion, where the kernel estimator provides a flexible, data-driven alternative to the parametric density. The kernel density estimator in the numerator is adaptable to both univariate ($d = 1$) and multivariate ($d \geq 2$) settings, as discussed by Silverman (1986) and Wand and Jones (1995). To the best of the authors' knowledge, the use of kernel density estimation in a likelihood ratio-type goodness-of-fit test has not been addressed in the existing literature.

The remainder of the paper is organized as follows. Section 2 reviews kernel density estimation for both univariate and multivariate data, including boundary correction techniques. Section 3 introduces the KDE-based test statistic. Section 4 describes its theoretical properties. Section 5 presents a bootstrap algorithm for calibrating the test. Section 6 reports simulation studies and real data applications to evaluate the performance of the proposed method. Section 7 concludes the paper with a discussion and potential directions for future research.

2 Kernel Density Estimation

KDE is a nonparametric technique for estimating a probability density function based on observed data. Given a random sample $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n) \in \mathbb{R}^d$ from an unknown density f , each $\mathbf{X}_i \in \mathbb{R}^d$ is typically treated as a row vector: $\mathbf{X}_i = (X_{i1}, X_{i2}, \dots, X_{id})$, where X_{ij} denotes the j -th coordinate of the i -th observation. When $d = 1$, i.e., in the univariate case, we drop the bold notation and write $X_1, \dots, X_n \in \mathbb{R}$ for simplicity.

The basic idea of KDE is to estimate f by averaging a kernel function centered at each data point. This section reviews the KDE methodology in both univariate and multivariate settings, with emphasis on practical issues such as bandwidth selection and boundary correction.

2.1 Univariate Kernel Density Estimation

Let $X = (X_1, \dots, X_n)$ be i.i.d. observations from a univariate continuous distribution with density f . A *kernel* is any smooth function K satisfying: The multivariate kernel function $K : \mathbb{R}^d \rightarrow \mathbb{R}$ is assumed to satisfy the following conditions:

$$K(u) \geq 0, \quad \int_{\mathbb{R}} K(u) du = 1, \quad \int_{\mathbb{R}} uK(u) du = \mathbf{0}, \quad \text{and} \quad \sigma_K^2 = \int_{\mathbb{R}} u^2 K(u) du < \infty.$$

(Wasserman, 2007). Common choices include:

- Gaussian kernel: $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$,
- Epanechnikov kernel: $K(u) = \frac{3}{4}(1 - u^2)\mathbb{I}(|u| \leq 1)$,
- Uniform kernel: $K(u) = \frac{1}{2}\mathbb{I}(|u| \leq 1)$,
- Biweight kernel: $K(u) = \frac{15}{16}(1 - u^2)^2\mathbb{I}(|u| \leq 1)$,

where $\mathbb{I}(\cdot)$ denotes the indicator function.

Given a kernel K and a positive number h (called the *bandwidth*), the kernel density estimator (KDE) is defined by

$$\hat{f}_n(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - X_i}{h}\right).$$

If f is continuous at x and $h_n \rightarrow 0$ with $nh_n \rightarrow \infty$ as $n \rightarrow \infty$, then $\hat{f}_n(x) \xrightarrow{p} f(x)$, where $\xrightarrow{p}$ denotes convergence in probability.

The choice of the bandwidth h is critical. A small h leads to high variance (undersmoothing), while a large h results in high bias (oversmoothing). One common approach to selecting h is cross-validation (Silverman, 1986; Wasserman, 2007). For the Gaussian kernel, the bandwidth minimizing the integrated mean squared error (IMSE) under normality is

$$h = \left(\frac{4}{3}\right)^{1/5} \hat{\sigma} n^{-1/5} \approx 1.06 \hat{\sigma} n^{-1/5}, \quad (3)$$

where $\hat{\sigma} = S$ is the sample standard deviation. This performs well for symmetric unimodal densities but tends to oversmooth skewed or multimodal ones. A more robust alternative uses:

$$\hat{\sigma} = \min(S, \text{IQR}/1.34), \quad (4)$$

where IQR is the interquartile range (Rizzo, 2000). Silverman (1986, p. 48) also proposes the following heuristic:

$$h = 0.9 \hat{\sigma} n^{-1/5} = 0.9 \min(S, \text{IQR}/1.34) n^{-1/5}. \quad (5)$$

For further methods, see Section 3.4 of Silverman (1986). No universally optimal method exists, and bandwidth selection remains context-dependent.

When the support of f is bounded, KDE may suffer from boundary bias. To address this, we use the *reflection method*, which symmetrizes the kernel window near boundaries (Jones, 1993; Silverman, 1986):

- **Left-bounded support** $[a, \infty)$:

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{i=1}^n \left\{ K\left(\frac{x - X_i}{h}\right) + K\left(\frac{x + X_i - 2a}{h}\right) \right\}, \quad x \geq a.$$

- **Right-bounded support** $(-\infty, b]$:

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{i=1}^n \left\{ K\left(\frac{x - X_i}{h}\right) + K\left(\frac{2b - x - X_i}{h}\right) \right\}, \quad x \leq b.$$

- **Bounded support** $[a, b]$:

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{i=1}^n \left\{ K\left(\frac{x - X_i}{h}\right) + K\left(\frac{x + X_i - 2a}{h}\right) + K\left(\frac{2b - x - X_i}{h}\right) \right\}, \quad x \in [a, b].$$

This approach is particularly effective with symmetric, rapidly decaying kernels like the Gaussian or Epanechnikov. It mitigates edge bias without distorting the estimator or introducing discontinuities.

2.2 Multivariate Kernel Density Estimation

Let $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)$ be i.i.d. observations from a d -dimensional continuous distribution with unknown density f . Each observation $\mathbf{X}_i \in \mathbb{R}^d$ is typically written as a row vector: $\mathbf{X}_i = (X_{i1}, \dots, X_{id})$, where X_{ij} denotes the j -th coordinate of the i -th observation. The multivariate kernel density estimator of f at a point $\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d$ is defined as

$$\hat{f}_n(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \frac{1}{|H|^{1/2}} K\left(H^{-1/2}(\mathbf{x} - \mathbf{X}_i)\right), \quad (6)$$

where $K : \mathbb{R}^d \rightarrow \mathbb{R}$ is a multivariate kernel function satisfying

$$K(\mathbf{u}) \geq 0, \quad \int_{\mathbb{R}^d} K(\mathbf{u}) d\mathbf{u} = 1, \quad \int_{\mathbb{R}^d} \mathbf{u} K(\mathbf{u}) d\mathbf{u} = \mathbf{0}, \quad \text{and} \quad \sigma_K^2 = \int_{\mathbb{R}^d} \|\mathbf{u}\|^2 K(\mathbf{u}) d\mathbf{u} < \infty.$$

Here, H is a $d \times d$ symmetric positive definite bandwidth matrix that governs the scale and orientation of smoothing. The determinant $|H|$ ensures proper normalization of the kernel, and $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^d$. For radial kernels (such as the multivariate Gaussian), the

argument $H^{-1/2}(\mathbf{x} - \mathbf{X}_i)$ standardizes the distance using H , and

$$\|H^{-1/2}(\mathbf{x} - \mathbf{X}_i)\|^2 = (\mathbf{x} - \mathbf{X}_i)^\top H^{-1}(\mathbf{x} - \mathbf{X}_i)$$

is the Mahalanobis distance.

In practice, a common simplification is the product kernel estimator:

$$\hat{f}_n(\mathbf{x}) = \frac{1}{n \prod_{j=1}^d h_j} \sum_{i=1}^n \prod_{j=1}^d K\left(\frac{x_j - X_{ij}}{h_j}\right), \quad (7)$$

where x_j and X_{ij} denote the j -th components of $\mathbf{x}$ and $\mathbf{X}_i$, respectively. This corresponds to using a diagonal bandwidth matrix $H = \text{diag}(h_1^2, \dots, h_d^2)$ and a product kernel of the form $K(\mathbf{u}) = \prod_{j=1}^d K(u_j)$.

For example, the multivariate Gaussian kernel is given by

$$K(\mathbf{u}) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\|\mathbf{u}\|^2\right),$$

and the multivariate Epanechnikov kernel is defined by

$$K(\mathbf{u}) = \frac{1}{c_d} (d+2)(1 - \|\mathbf{u}\|^2) \mathbb{I}(\|\mathbf{u}\| \leq 1),$$

where $c_d = \frac{2\pi^{d/2}}{d\Gamma(d/2)}$ is the volume of the d -dimensional unit ball. When $d = 1$, this reduces to the univariate Epanechnikov kernel with $c_1 = 2$.

In the product kernel setting, a commonly used rule-of-thumb for multivariate normal data is

$$h_j = \left(\frac{4}{d+2}\right)^{1/(d+4)} \hat{\sigma}_j n^{-1/(d+4)}, \quad (8)$$

where $\hat{\sigma}_j$ is the sample standard deviation of the j -th component. Scott's rule (Scott, 1992) simplifies this to

$$h_j = \hat{\sigma}_j n^{-1/(d+4)}.$$

As in the univariate case, several data-driven methods can be used to select the bandwidth matrix H (Scott, 1992; Wand and Jones, 1995). However, boundary correction in the multivariate case is significantly more complicated and is often omitted, particularly when the support is unbounded or ill-defined.

3 Test Statistic and Asymptotic Results

This section introduces a family of goodness-of-fit test statistics based on KDE, defined in (2). The proposed statistic compares a nonparametric estimate of the density with a parametric model specified under the null hypothesis. We show that the normalized log-statistic converges to zero under the null and to a strictly positive constant under fixed alternatives, confirming the consistency of the test.

Let $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n) \in \mathbb{R}^d$ be a random sample from an unknown continuous density f . We aim to test the null hypothesis given in (1). Let $\hat{\theta}$ denote the maximum likelihood estimator under H_0 , and let $\hat{f}_n$ denote the kernel density estimator defined in Section 2. For numerical stability and interpretability we typically work of the logarithmic form of (2) given by. Thus, the test statistic is

For numerical stability and ease of interpretation, we typically work with the logarithmic form of (2), given by

$$\log T_n = \sum_{i=1}^n \left(\log \hat{f}_n(\mathbf{X}_i) - \log f_0(\mathbf{X}_i; \hat{\theta}) \right). \quad (9)$$

Large values of $\log T_n$ indicate divergence from the model and support rejection of the null hypothesis.

In the univariate case ($d = 1$), if the support of f is bounded, we use a boundary-corrected KDE (e.g., the reflection method from Section 3.1). In the multivariate case, we use a general bandwidth matrix H . When a spherically symmetric kernel (e.g., Gaussian) and a scalar bandwidth matrix are used, the statistic is invariant under orthogonal transformations of the data—a desirable property in multivariate analysis (Wand and Jones, 1995).

We assume the following regularity conditions:

- (C1) The true density f is continuous and strictly positive on its support in $\mathbb{R}^d$.
- (C2) The parametric family $f_0(\mathbf{x}; \theta)$ is continuous and strictly positive on the support of f , and for all $\theta \in \Theta$, the function $\log f_0(\mathbf{x}; \theta)$ is integrable with respect to f .
- (C3) The KDE $\hat{f}_n(\mathbf{x})$ is pointwise consistent for $f(\mathbf{x})$ as $n \rightarrow \infty$, i.e., for each fixed $\mathbf{x}$, $\hat{f}_n(\mathbf{x}) \xrightarrow{P} f(\mathbf{x})$.

Theorem 1 *Under conditions (C1)–(C3), the following properties hold for the log-statistic defined in (9):*

- (i) *Under $H_0 : f = f_0(\cdot; \theta_0)$ for some $\theta_0 \in \Theta$,*

$$\frac{1}{n} \log T_n \xrightarrow{P} 0.$$

- (ii) *Under a fixed alternative $H_1 : f \neq f_0(\cdot; \theta)$ for all $\theta \in \Theta$,*

$$\frac{1}{n} \log T_n \xrightarrow{P} \mathbb{E}_f \left[\log \frac{f(\mathbf{X})}{f_0(\mathbf{X}; \theta^*)} \right] > 0,$$

where θ^* minimizes the Kullback–Leibler divergence

$$\text{KL}(f \parallel f_0(\cdot; \theta)) = \int_{\mathbb{R}^d} f(\mathbf{x}) \log \frac{f(\mathbf{x})}{f_0(\mathbf{x}; \theta)} d\mathbf{x}$$

over $\theta \in \Theta$ (White, 1982).

Proof. We write

$$\frac{1}{n} \log T_n = \frac{1}{n} \sum_{i=1}^n \left(\log \hat{f}_n(\mathbf{X}_i) - \log f_0(\mathbf{X}_i; \hat{\theta}) \right).$$

- (i) Under H_0 , we have $f = f_0(\cdot; \theta_0)$ for some $\theta_0 \in \Theta$. By (C3), $\hat{f}_n(\mathbf{X}_i) \xrightarrow{P} f_0(\mathbf{X}_i; \theta_0)$, and hence by the continuous mapping theorem we have

$$\log \hat{f}_n(\mathbf{X}_i) \xrightarrow{P} \log f_0(\mathbf{X}_i; \theta_0).$$

Also, $\hat{\theta} \xrightarrow{P} \theta_0$, so $\log f_0(\mathbf{X}_i; \hat{\theta}) \xrightarrow{P} \log f_0(\mathbf{X}_i; \theta_0)$. Thus, each term in the sum converges to zero, and the weak law of large numbers gives

$$\frac{1}{n} \log T_n \xrightarrow{P} 0.$$

(ii) Under H_1 , we assume that $f \neq f_0(\cdot; \theta)$ for all $\theta \in \Theta$. Then by standard MLE theory, $\hat{\theta} \xrightarrow{P} \theta^*$, where θ^* is the minimizer of $\text{KL}(f \| f_0(\cdot; \theta))$ over $\theta \in \Theta$ (We need a reference/citation here). Meanwhile, the KDE satisfies $\hat{f}_n(\mathbf{X}_i) \xrightarrow{P} f(\mathbf{X}_i)$ by pointwise consistency. Therefore,

$$\log \hat{f}_n(\mathbf{X}_i) - \log f_0(\mathbf{X}_i; \hat{\theta}) \xrightarrow{P} \log f(\mathbf{X}_i) - \log f_0(\mathbf{X}_i; \theta^*).$$

Since $\log(f(\mathbf{X}_i)/f_0(\mathbf{X}_i; \theta^*))$ is integrable by condition (C2), the weak law of large numbers yields

$$\frac{1}{n} \log T_n \xrightarrow{P} \mathbb{E}_f \left[\log \frac{f(\mathbf{X})}{f_0(\mathbf{X}; \theta^*)} \right],$$

which is strictly positive by the identifiability of f_0 and the assumption that $f \neq f_0(\cdot; \theta^*)$ almost everywhere. ■

4 Bootstrap Calibration

As established in the previous section, the KDE-based test statistic exhibits desirable asymptotic behavior under both the null and alternative hypotheses. However, its finite-sample distribution under H_0 is analytically intractable due to the presence of a nonparametric density estimator in the numerator. To address this, we employ a parametric bootstrap procedure to approximate the null distribution and obtain p -values.

The following algorithm provides a unified bootstrap calibration procedure applicable to both univariate and multivariate settings. For $d = 1$, boundary correction methods (e.g., reflection) are applied to the kernel density estimator. For $d \geq 2$, a general bandwidth matrix is used for multivariate KDE.

Bootstrap Calibration Algorithm

Step 1. Compute the observed test statistic.

Given observations $\{\mathbf{X}_1, \dots, \mathbf{X}_n\} \subset \mathbb{R}^d$, compute $\log T_n$ as defined in (9). For the univariate case ($d = 1$), use boundary correction as needed. For the multivariate case ($d \geq 2$), use a general bandwidth matrix H in constructing the KDE.

Step 2. Generate bootstrap samples under H_0 .

For $b = 1, \dots, B$:

- (a) Sample $\{\mathbf{X}_1^{*(b)}, \dots, \mathbf{X}_n^{*(b)}\}$ i.i.d. from $f_0(\cdot; \hat{\theta})$.
- (b) Compute the KDE $\hat{f}_n^{*(b)}$ using the bootstrap sample, with the same kernel and bandwidth method used in Step 1.
- (c) Compute the bootstrap statistic:

$$\log T_n^{*(b)} = \sum_{i=1}^n \left(\log \hat{f}_n^{*(b)}(\mathbf{X}_i^{*(b)}) - \log f_0(\mathbf{X}_i^{*(b)}; \hat{\theta}) \right),$$

Step 3. Compute the bootstrap p -value.

$$p_{\text{boot}} = \frac{1}{B} \sum_{b=1}^B \mathbb{I} \left(T_n^{*(b)} \geq T_n \right).$$

Step 4. Make a decision.

Reject H_0 at significance level α if $p_{\text{boot}} < \alpha$.

5 Simulation Study

In this section, we evaluate the empirical performance of the proposed KDE-based goodness-of-fit test through a series of simulation studies. Our objectives are to assess the control of Type I error under the null hypothesis and to examine the power of the test under various alternatives. Both univariate and multivariate settings are considered.

For each scenario in Examples 1–3, we generate $M = 1000$ independent samples of size $n \in \{30, 50, 100, 200\}$ from known distributions. In all examples, the corresponding p -values are obtained using the bootstrap algorithm described in Section 4 with $B = 500$ bootstrap replicates. For kernel estimation, different kernels are used for comparison purposes. However, for the bandwidth, we adopt the well-known rule of thumb given in (3).

5.1 Univariate Examples

Example 1. Testing Normality: In this example, we consider the null hypothesis $H_0 : f = f_0$, where f_0 is the density of $\mathcal{N}(\mu, 1)$, with μ unknown and estimated by the sample mean $\hat{\mu} = \bar{x}$. To examine the sensitivity of the proposed KDE-based test, we generate data from a variety of alternative distributions, each deviating from normality in a specific way:

- t_3 distribution (heavy tails),
- Logistic(0,1) distribution (different kurtosis),
- Cauchy(0,1) distribution (infinite variance),
- Mixture: $0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$ (bimodality),
- $\mathcal{N}(0, 4)$ (variance misspecification).

For each distribution, we simulate $M = 1000$ independent datasets of sizes $n \in \{30, 50, 100, 200\}$. The kernel density estimator in the test statistic is computed using various kernels for comparison, while the bandwidth h is selected using the rule of thumb in (5). Table 1 reports the empirical rejection rates at the $\alpha = 0.05$ level based on the bootstrap calibration procedure described in Section 4.

Example 2. Testing for Exponentiality (Boundary Case): In this example, we test the null hypothesis $H_0 : f = f_0$, where f_0 is the density of an exponential distribution with unknown mean λ , estimated by the sample mean $\hat{\lambda} = \bar{x}$ for each sample. To evaluate the sensitivity of the proposed test, we generate samples from several alternative distributions supported on $[0, \infty)$:

Table 1: Empirical rejection rates for the KDE-based test in Example 1. Results are based on $M = 1000$ replications with significance level $\alpha = 0.05$.

Distribution	n	h	Gaussian(h)	Epanechnikov(h)	Uniform(h)	Biweight(h)
$\mathcal{N}(0, 1)$	30	0.412353	0.045	0.041	0.044	0.042
	50	0.3846479	0.048	0.052	0.055	0.059
	100	0.3432996	0.048	0.051	0.055	0.051
	200	0.302599	0.060	0.064	0.062	0.053
t_3	30	0.4957008	0.757	0.763	0.791	0.736
	50	0.4570646	0.870	0.869	0.907	0.861
	100	0.4040209	0.993	0.990	0.991	0.983
	200	0.3571162	0.999	0.999	1.000	1.000
Logistic(0,1)	30	0.6991075	0.958	0.957	0.970	0.949
	50	0.6445828	0.995	0.996	0.999	0.994
	100	0.5733677	1	1	1	1
	200	0.5076416	1	1	1	1
Cauchy(0,1)	30	0.6980617	0.999	0.999	0.999	0.998
	50	0.6271982	1	1	1	1
	100	0.5410886	1	1	1	1
	200	0.4669609	1	1	1	1
Mixture	30	0.6155031	0.692	0.696	0.794	0.703
	50	0.5694524	0.877	0.884	0.936	0.880
	100	0.5002439	0.986	0.992	0.999	0.987
	200	0.4404969	1	1	1	1
$\mathcal{N}(0, 4)$	30	1.649412	1	1	1	1
	50	1.538592	1	1	1	1
	100	1.373198	1	1	1	1
	200	1.210396	1	1	1	1

Table 2: Empirical rejection rates for the KDE-based test in Example 2. Results are based on $M = 1000$ replications, with significance level $\alpha = 0.05$.

Scenario	n	h	Gaussian	Epanechnikov	Uniform	Biweight
Exp(1)	30	0.3507307	0.024	0.02	0.022	0.018
	50	0.3260124	0.030	0.038	0.036	0.044
	100	0.2897109	0.042	0.043	0.039	0.039
	200	0.2533352	0.050	0.046	0.043	0.043
Gamma(2,1)	30	0.5439781	0.178	0.244	0.515	0.376
	50	0.5032442	0.438	0.558	0.860	0.693
	100	0.4502891	0.923	0.960	1	0.984
	200	0.3974645	1	1	1	1
Weibull(1.5,1)	30	0.2486300	0.139	0.195	0.464	0.304
	50	0.2295365	0.354	0.468	0.800	0.631
	100	0.2054564	0.831	0.911	0.989	0.955
	200	0.1827772	0.998	0.999	1	0.999
Lognormal(0,1)	30	0.4774330	0.304	0.321	0.304	0.322
	50	0.4319098	0.517	0.540	0.509	0.524
	100	0.3823509	0.871	0.882	0.844	0.856
	200	0.3378424	0.999	0.998	0.999	0.998

- Gamma(2, 1),
- Weibull(1.5, 1),
- Lognormal(0, 1).

To address the boundary bias inherent in kernel density estimation on the non-negative real line, we apply the reflection method described in (??) with $a = 0$. Table 4 reports the empirical rejection rates based on $M = 1000$ simulations at the $\alpha = 0.05$ significance level.

5.2 Multivariate Examples

Example 3. Testing Multivariate Normality: We now consider the multivariate case with dimension $d = 2$, and test the null hypothesis $H_0 : f = f_0$, where $f_0 = \phi_2(\cdot; \mathbf{0}, I_2)$ is the bivariate normal density with mean $\mathbf{0}$ and identity covariance matrix I_2 . For each simulated sample, the KDE-based test is applied using two multivariate kernels: Gaussian and Epanechnikov. The estimator $\hat{f}_n(\mathbf{x})$ is computed as in (7), using a diagonal bandwidth matrix with entries h_j as defined

Table 3: Empirical rejection rates for the KDE-based test in Example 3. Results are based on $M = 1000$ replications, with significance level $\alpha = 0.05$.

Scenario	$h = (h_1, h_2)$	n	Gaussian	Epanechnikov
$\mathcal{N}(\mathbf{0}, I_2)$	(0.5635, 0.5612)	30	0.054	0.048
	(0.5200, 0.5167)	50	0.067	0.057
	(0.4610, 0.4612)	100	0.059	0.059
	(0.4139, 0.4129)	200	0.05	0.043
Bivariate t_3	(0.9141, 0.9162)	30	0.91	0.87
	(0.8606, 0.8745)	50	0.98	0.97
	(0.7698, 0.7783)	100	1	1
	(0.6989, 0.6926)	200	1	1
Contaminated Normal	(0.7633, 0.7588)	30	0.864	0.812
	(0.6960, 0.6964)	50	0.964	0.934
	(0.6220, 0.6234)	100	0.999	0.995
	(0.5559, 0.5571)	200	1	1
Uniform on $[0, 1]^2$	(0.1636, 0.1632)	30	1	1
	(0.1502, 0.1496)	50	1	1
	(0.1340, 0.1337)	100	1	1
	(0.1190, 0.1995)	200	1	1

in (8). To evaluate the sensitivity of the test, we generate data from the following alternative distributions:

- Bivariate t_3 distribution (heavy tails),
- Contaminated normal: $0.9\mathcal{N}(\mathbf{0}, I_2) + 0.1\mathcal{N}(\mathbf{3}, I_2)$ (bimodal structure),
- Uniform distribution on $[0, 1]^2$ (non-elliptical support).

Table 3 reports the empirical rejection rates based on $M = 1000$ replications, with a significance level of $\alpha = 0.05$.

5.3 Real Data Applications

In this section, we evaluate the practical performance of the proposed KDE-based goodness-of-fit test using real datasets. Both univariate and multivariate examples are considered.

Example 4. *Testing a Laplace Model (Yarn Breaking Strength Data):* We consider the classical

dataset of breaking strength measurements for 100 yarns, originally reported by Duncan (1974). The observations are univariate and continuous. Previous studies have suggested that the Laplace distribution provides a reasonable fit.

We test the null hypothesis $H_0: f = f_0$, where f_0 is the density of the Laplace(μ, σ) distribution, given by

$$f_0(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x - \mu|}{\sigma}\right), \quad x \in \mathbb{R}.$$

The parameters are estimated via maximum likelihood, yielding $\hat{\mu} = 99$ and $\hat{\sigma} = 8.33$. The KDE-based test is applied using a Gaussian kernel with bandwidth h selected as in (5). A bootstrap p -value is computed using $B = 1000$ replicates.

The resulting bootstrap p -value is $p_{\text{boot}} = 0.1520$, indicating no evidence against the Laplace model. This supports earlier findings and confirms the adequacy of the parametric fit.

Example 5. *Testing a Gamma Model (River Flow Data):* This example analyzes river flow measurements (in millions of cubic feet per second) from the Susquehanna River at Harrisburg, Pennsylvania, recorded over the five-year period 1980–1984. The dataset is as follows:

0.654, 0.613, 0.315, 0.449, 0.297, 0.402, 0.379, 0.423, 0.379, 0.324,
0.269, 0.740, 0.418, 0.412, 0.494, 0.416, 0.338, 0.392, 0.484, 0.265

We assess whether a three-parameter Gamma distribution provides an adequate fit. That is, we test $H_0: f = f_0$, where f_0 denotes the Gamma(α, β, γ) density:

$$f(x; \alpha, \beta, \gamma) = \frac{\beta^{-\alpha}(x - \gamma)^{\alpha-1}e^{-\beta(x-\gamma)}}{\Gamma(\alpha)}, \quad x > \gamma,$$

where $\alpha > 0$, $\beta > 0$, and $\gamma > 0$ are the shape, rate, and location parameters, respectively.

Maximum likelihood estimates yield $\hat{\alpha} = 1.705$, $\hat{\beta} = 0.0955$, and $\hat{\gamma} = 0.2602$. Since the data are supported on $[0, \infty)$, boundary bias is corrected using the reflection method as described in left bound support with $a = 0$, and h is selected using (5). The resulting bootstrap p -value is given in Table 4, providing strong

Table 4: p-values of the KDE-based test in Example 5 based on different kernels.

h	Gaussian	Epanechnikov	Uniform	Biweight
0.045	0.538	0.581	0.376	0.617

support for the Gamma model. This example highlights the importance of boundary correction in KDE when analyzing positive-valued data.

Example 6. *Testing Multivariate Normality (Iris Dataset):* We consider the classical Iris dataset, which includes measurements on sepal length, sepal width, petal length, and petal width for three iris species. We restrict attention to the 50 observations corresponding to the Setosa species, and test whether the bivariate vector of petal length and petal width follows a bivariate normal distribution.

We test $H_0: f = f_0$, where f_0 is the density of a bivariate normal distribution with unknown mean and covariance matrix. Parameters are estimated via maximum likelihood. The KDE-based test is applied using both Gaussian and Epanechnikov multivariate kernels. The estimator $\hat{f}_n(\mathbf{x})$ is computed as in (7), using a diagonal bandwidth matrix with entries h_j determined by (8).

The resulting bootstrap p -value is $p_{\text{boot}} = 0.0220$, indicating moderate evidence against the bivariate normal assumption. This suggests that the petal measurements may exhibit deviations from normality, such as skewness or heavy tails.

6 Conclusion

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